Alexi

Intelligent LLM Orchestrator for SAP AI Core

v0.2.6

The Challenge

1. **Fragmented Access**

   LLMs scattered across multiple providers with different APIs and authentication

2. **No Smart Routing**

   Manual model selection — no intelligent matching of prompts to the right model

3. **Missing Safeguards**

   No enterprise-grade content filtering, PII masking, or compliance controls

4. **Context Chaos**

   Manual session management, no persistent memory, no cost tracking

Enterprise AI Today

Multiple APIs
Multiple credentials
No unified control
No intelligent routing

What is Alexi?

CLI

Unified CLI Tool

Available as alexi or ax — a single command-line interface to SAP AI Core's orchestration service

AI

Intelligent Auto-Routing

Classifies every prompt by task type and complexity, then selects the optimal model automatically

25+

Autonomous Coding Agent

25+ built-in tools for file operations, bash execution, web access, code search, and memory

Key Capabilities

Auto-Routing

Classifies prompts, scores models, picks the best fit automatically

Agentic Tool Loop

Autonomous multi-step task execution with tool calls

25+ Built-in Tools

File ops, bash, web fetch, search, memory, browser

Multi-Model Support

GPT, Claude, Gemini, Llama, Nova, DeepSeek, Mistral

MCP Integration

Client + Server for external tool extensibility

Enterprise Features

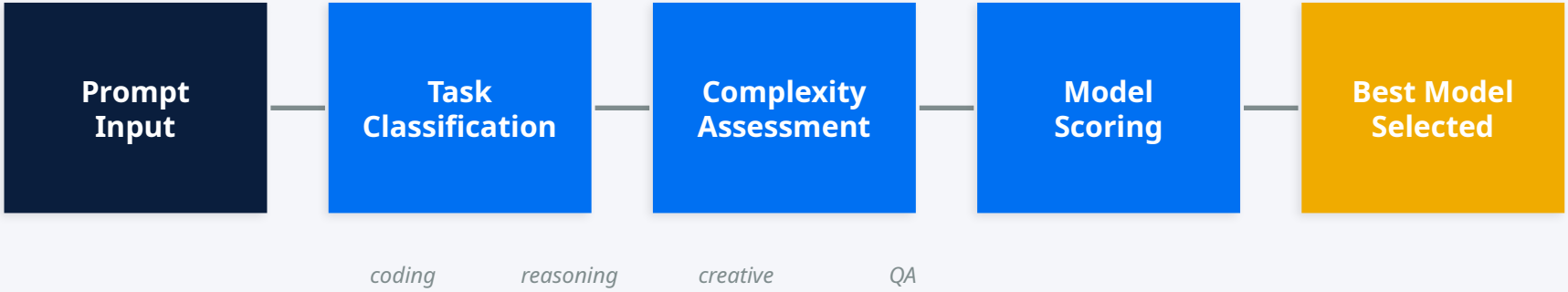
Content filtering, PII masking, grounding, translation

Supported Models

OpenAI	Anthropic	Google	Amazon
GPT-4o GPT-4.1 GPT-5	Claude 4.5 Opus Claude Sonnet Claude Haiku	Gemini	Nova

Meta	DeepSeek	Mistral	SAP
Llama	DeepSeek	Mistral	ABAP

Intelligent Auto-Routing



- Hot-reloadable routing config

--prefer-cheap flag for cost optimization

Architecture Overview

CLI Layer	Commander + Ink/React TUI
Core Layer	Orchestrator, Router, Session Mgr, Memory, Cost Tracker
Agent Layer	Code, Debug, Plan, Explore, Orchestrator Agents
Tool Layer	25 Built-in Tools + MCP Tools
Provider Layer	SAP Orchestration Provider
SAP AI Core	Content Filter, DPI, Grounding, Translation
LLM Models	GPT, Claude, Gemini, Llama, Nova, DeepSeek, Mistral

Layered architecture with clean separation of concerns

Agent System

Agent	Role	Tool Access
Code	General coding, all tool access	Full (25+ tools)
Debug	Bug fixing, error analysis	Full (25+ tools)
Plan	Architecture, read-only analysis	read, glob, grep, webfetch
Explore	Fast codebase search	read, glob, grep
Orchestrator	Delegates to sub-agents	task only

@syntax switching: @debug fix this error — instantly switches agent context
Each agent has custom temperature, system prompt, and tool permissions

Tool Ecosystem

File Operations

read, write, edit, multiedit, delete, glob, grep, ls

Execution

bash

Web

webfetch, websearch, browser

Agent / Workflow

task, question, todowrite, skill

Intelligence

definitions, codesearch, warpgrep, diagnostics, storeMemory, recallMemory, batch, notebookRead/Edit

SAP Enterprise Features

Content Filtering

Azure Content Safety + Llama Guard 3
Input and output filtering for safe, compliant AI interactions

Data Masking (DPI)

Anonymize 25+ PII entity types
Email, phone, SSN, credit cards, addresses, and more

Document Grounding

Vector repositories + help.sap.com knowledge bases for context-aware responses

Translation

Input/output language translation
Seamless multilingual support for global teams

System Prompt Pipeline

1	**Soul**	Core identity and behavior directives
2	**Model-Specific**	Tailored for Anthropic / OpenAI / Gemini
3	**Environment**	Working dir, git status, platform, date
4	**Agent Role**	Code / Debug / Plan / Explore / Orchestrator
5	**Instruction Files**	AGENTS.md, user rules, project rules
6	**Custom Rules**	User-provided via --system flag

Cache-friendly ordering: stable content first for LLM prompt caching

MCP Integration

MCP Server

Exposes all 25+ tools via stdio transport

External AI clients (VS Code, Cursor, other agents) can use Alexi's tools directly

MCP Client

Connects to external MCP servers

Filesystem, GitHub, Brave Search — aggregates tools from any MCP server

Config: ~/.alexi/mcp-servers.json

Terminal UI

```
$ alexi -i

Alexi v0.2.6 | Model: gpt-4o

> Refactor the auth module

[read] src/auth/index.ts
[edit] src/auth/middleware.ts
[bash] npm test

All tests passing. Cost: $0.023
```

- Message bubbles with markdown rendering
- Diff view for code changes
- Tool call visualization blocks
- Command palette with slash commands
- Spinner animations + syntax highlighting

Built with Ink v6 + React 19

Cost & Session Management

Cost Tracking

- Per-model pricing tables
- Per-session / per-day / per-month summaries
- CSV export capability
- Real-time cost display during sessions

Session Management

- Persistent multi-turn conversations
- Context compaction at 90% window
- Typed memory system:
  - episodic, semantic, procedural, working

Demo Flow

01	**Launch**	alexi -i (interactive mode)
02	**Auto-route**	Send a prompt, watch model selection
03	**Agent Execution**	Multi-step coding task with tool calls
04	**Tool Visualization**	See file reads, edits, bash execution
05	**Cost Summary**	Review session costs and model usage

This slide outlines what to demo live

Tech Stack

TypeScript 5.9

Strict mode

Node.js 22+

ES Modules

SAP AI SDK

Orchestration + AI API

Ink v6 + React 19

Terminal UI

Zod v4

Runtime validation

Commander 14

CLI framework

Hono v4

HTTP server

Tree-sitter

Code parsing

Vitest 4

Testing framework

Thank You

Questions?

Alexi CLI — alexi / ax
SAP AI Core Orchestration